Fiscal Year:	FY 2024	**Task Last Updated:**	FY 07/30/2024
PI Name:	Huang, Alex M.D., Ph.D.		
Project Title:	Exercise Countermeasure to Prevent Ocular Structural and Functional Changes in a Terrestrial Model of SANS (80NSSC22K1803)		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **SANS**:Risk of Spaceflight Associated Neuro-ocular Syndrome (SANS)		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92093-0946	**Congressional District:**	52
Comments:	The PI moved from Doheny Eye Institute to University of California, San Diego in 2023.		
Project Type:	Ground	**Solicitation / Funding Source:**	2018-2019 HERO 80JSC018N0001-SANS: Spaceflight Associated Neuro-ocular Syndrome Countermeasures. Appendix C
Start Date:	08/22/2022	**End Date:**	08/21/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	7
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	8
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Laurie, Steven (KBR./NASA Johnson Space Center) Lee, Stuart (KBR/NASA Johnson Space Center) Macias, Brandon (NASA Johnson Space Center) Marshall-Goebel, Karina (KBR/NASA Johnson Space Center) Sadda, Srinivas (Doheny Eye Institute) Linda, Loerch (NASA Johnson Space Center) Young, Millennia (NASA Johnson Space Center)		
Grant/Contract No.:	80NSSC22K1803		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Ed. Note (3/30/23): Project continues "Exercise Countermeasure to Prevent Ocular Structural and Functional Changes in a Terrestrial Model of SANS," Grant #80NSSC20K1034 with original Period of Performance 07/01/2020-06/30/2023. See that project for previous reporting. Optic disc edema develops in about 16% of astronauts during long-duration spaceflight and is of high concern to the NASA medical community and a target of therapeutic treatment given risk of vision loss. Currently, there is no reliable method to predict which crewmembers will develop disc edema. Moreover, it was previously believed that only the spaceflight environment could produce optic disc edema in normal healthy subjects. Recently our research team developed a novel ground-based spaceflight analog that reproduced disc edema in healthy test subjects that can be used to test novel pathophysiological hypotheses and possible Spaceflight Associated Neuro-ocular Syndrome (SANS) countermeasures. The research outlined in this proposal will use this new ground-based spaceflight analog advanced by NASA to elucidate the structural and functional impact of optic disc edema and to evaluate alterations to retinal and optic nerve blood flow to understand their contributions to the etiology of SANS. Further, a novel countermeasure will be tested to prevent the development of disc edema, functional decline in ganglion cell function, and vascular alterations associated with this ground-based spaceflight analog. This proposal will (1) determine combined structural and functional ocular alterations caused by the development of optic disc edema in this spaceflight analog, (2) determine the role of altered vascular blood flow in the development of optic disc edema in this spaceflight analog, and (3) determine if daily aerobic exercise in combination with veno-occlusive thigh cuffs can be used as a preventative countermeasure against the formation of optic disc edema. Thus, this proposal utilizes a newly developed spaceflight analog and will allow us to test novel hypotheses for predicting, characterizing, and preventing the development of optic disc edema.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Outcomes of this research can benefit humans in space and on Earth. First, SANS is a hurdle to long-haul space travel. The ocular alterations in SANS can impact vision, which represents a major safety concern during spaceflight. Thus, better understanding SANS and developing countermeasures to mitigate SANS improves the safety of space travel. Then, the major pathophysiological hypothesis for why SANS is occurring is the idea of fluid shifts. This concept states that gravity normally pulls total body fluid to the lower extremities of humans, and without gravity this fluid re-distributes to the head. It is hypothesized that the increased volume in the head leads to SANS. Increased volume in the head can also lead to other clinical manifestations such as a sensation of facial fullness and increased intraocular (eye) or intracranial (brain) pressure. The latter two are important in many Earth-bound diseases. For example, in glaucoma, head-ward fluid shifts are known to increase eye pressure which can lead to blindness. Thus, better understanding fluid shifts and how to minimize them may lead to additional benefits in the treatment of Earth-bound diseases.
Task Progress:	The data acquisition component of this study was completed in collaboration with the :envihab facility in the German Aerospace Center (DLR). Four campaigns were completed, and the Aims, Methods, and preliminary Results of this proposal are described below: Specific Aims Specific Aim 1: To determine if donning veno-occlusive thigh cuffs during and for 120 minutes after daily aerobic exercise can be used to (a) reverse headward fluid shift from bedrest for a sustained period of time and (b) prevent ocular structural and functional changes from developing at the end of bedrest. Hypothesis 1a: Donning veno-occlusive thigh cuffs during and after daily aerobic exercise will provide an acute and sustained fluid redistribution into the lower extremities (measured using femoral artery conductance and vasculature cross-sectional area) with a corresponding reduction in internal jugular vein cross-sectional area for up to 2 hours after cessation of exercise. Hypothesis 1b: There will be a decrease in prevalence and magnitude of optic disc edema in subjects performing the exercise Countermeasure (CM) outlined in SA1 compared to subjects in the head-down tilt group who perform no CM. Advanced analyses of optical coherence tomography (OCT) images of the optic nerve head combined with electrophysiological assessment of ganglion cell function will unveil functional decline and together may serve as a more sensitive endpoint compared to the visual fields tests currently used in astronauts. Specific Aim 2: To determine the role of altered vasculature in the development of optic disc edema in this spaceflight analog. Hypothesis 2a: Optical coherence tomography angiography (OCTA) will reveal vascular alterations in foveal avascular zone (FAZ) area or vascular density in the macula and nerve with intravenous fluorescein angiography showing leakage in subjects demonstrating the greatest degree of retinal thickening. Hypothesis 2b: There will be a decrease in OCTA and intravenous fluorescein angiography alterations in subjects performing exercise CM outlined in SA1 compared to subjects in the head-down tilt group who perform no CM. Materials and Methods This study occurred at the :envihab facility in the German Aerospace Center (DLR). This study was a part of a larger head-down tilt bedrest (HDTBR) CM study. There were 4 campaigns and 4 conditions. The four arms of the study included (1) a 6-hour upright control and (2) a 30-day bedrest 6-degrees head-down tilt (HDT) control. All experimental arms included the 30-day 6-degrees bedrest HDT condition combined with two CMs: (1) veno-occlusive thigh cuffs and exercise and (2) lower-body negative pressure (LBNP). The CM focus of this proposal is the exercise and cuff CM. For this proposal, the exercise and veno-occlusive thigh cuff CM were performed 6 days a week. Each session included exercise (60 min) that was sustained at ~60% of pre-bedrest peak oxygen consumption with the veno-occlusive thigh cuff applied at 50 mm Hg for six hours after exercise cessation. This length of time was increased over the original plan to better match inflight data. Endpoints were measured before, during, and after the 30-day bedrest period. These included intraocular pressure (IOP), optical coherence tomography (Spectralis HRA+OCT) of the macula and optic nerve, ultrasound measure of neck and lower extremity vessels, optical coherence tomography angiography (OCTA) of the macula and optic nerve, and electrophysiology for ganglion cell function. Intravenous fluorescein angiography (IVFA) was performed to assess blood:brain vascular integrity at the optic nerve head immediately after cessation of the 30-day bedrest period. Overall, OCTA, IVFA, and ERG endpoints will be assessed across all arms in addition to subjects in the exercise and cuff CM group. Results This study occurred at the :envihab facility at the German Aerospace Center (DLR). There were 4 campaigns. Campaigns 1 and 2 (C1/C2) were comprised of 12 subjects each. For each group of 12, 6 subjects received the upright CM, and 6 subjects received the LBNP CM. C1 ended in 12/2021 with data acquired from 12 subjects. C2 finished in

	3/2022 with data acquired from 11 subjects (5 upright CM and 6 LBNP CM). One subject (in the upright CM group) was removed from the study due to failure to follow instructions. Campaigns 3 and 4 (C3 and C4) were also comprised of 12 subjects each. For each group of 12, 6 subjects received the cuff and exercise CM, and 6 subjects served as controls who only underwent 30-days of strict 6-degrees head-down tilt HDTBR. C3 and C4 began in 1/2023. C3 was completed on 4/2023 for all 12 subjects. C4 was completed on 7/2023 for all 12 subjects. Data analysis is underway for all campaigns. Preliminarily, optic nerve head edema was seen in some cases, consistent with prior reports using HDTBR. This was observed by taking a picture of the nerve and by using OCT. IVFA then demonstrated that in some cases the nerves leaked, indicative of potential damage to the blood:ocular barrier and consistent with optic nerve head swelling. Currently, these endpoints as well as the others described above are being quantitated for comparison between groups to identify if any of the CM were effective. Complete List of All Abstract Presentations (1) Huang, A.S.: Spaceflight Associated Neuro-ocular Syndrome: SANS is a VIP. The inaugural John B Charles Memorial Lecture. NASA Human Research Program Investigators Workshop. Galveston, TX, 2/2024. Keynote Lecture (2) Huang, A.S.: SANS Research Direction and Novel Hypotheses. North American Neuro-Ophthalmologic Society. Honolulu, HI, 3/2024. Paper Presentation. (3) Huang, A.S., Karanjia, R., He, Y., Laurie, S.S., Poczatek, M., Lee, S.M.C., Young, M., Marion, K., Sadda, S., Macias, B.R. Exercise and Lower Extremity Thigh Cuff Countermeasures to Ocular Changes During Strict Head-down Tilt Bedrest: The SANS Countermeasure Bedrest Study. NASA Human Research Program Investigators Workshop. Galveston, TX, 2/2024. Paper Presentation. (4) Wanderer, D., Zhang, X., Ye, H., Karanjia, R., Sadda, S., Macias, B., Laurie, S., and Huang, A.S. Optic Nerve Vascular leaking in a 30-day Head-down Tilt Bedrest Model of Space-flight Associated Neuro-ocular Syndrome (SANS). ARVO. Seattle, WA. 5/2/24. Poster Presentation. (5) Christopher, M., Jalil, J., Weinreb, R.N., Laurie, S., Macias, B., and Huang, A.S. Predicting Spaceflight Associated Neuro-ocular Syndrome Using Artificial Intelligence. ARVO. Seattle, WA. 5/6/24. Poster Presentation.
Bibliography Type:	Description: (Last Updated: 07/02/2026)
Abstracts for Journals and Proceedings	Huang, AS. "Spaceflight associated neuro-ocular syndrome: SANS is a VIP. The inaugural John B Charles Memorial Lecture." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024
Abstracts for Journals and Proceedings	Huang, AS. "SANS research direction and novel hypotheses." 2024 North American Neuro-Ophthalmologic Society Annual Meeting, Honolulu, Hawaii, March 2-7, 2024. Abstracts. 2024 North American Neuro-Ophthalmologic Society Annual Meeting, Honolulu, Hawaii, March 2-7, 2024. , Mar-2024
Abstracts for Journals and Proceedings	Huang AS, Karanjia R, He Y, Laurie SS, Poczatek M, Lee SMC, Young M, Marion K, Sadda S, Macias BR. "Exercise and lower extremity thigh cuff countermeasures to ocular changes during strict head-down tilt bedrest: The SANS Countermeasure Bedrest Study." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024
Abstracts for Journals and Proceedings	Wanderer D, Zhang X, Ye H, Karanjia R, Sadda S, Macias BR, Laurie SS, and Huang AS. "Optic Nerve Vascular leaking in a 30-day Head-down Tilt Bedrest Model of Space-flight Associated Neuro-ocular Syndrome (SANS)" 2024 ARVO Annual Meeting, Seattle, Washington, May 5-9, 2024. Abstracts. 2024 ARVO Annual Meeting, Seattle, Washington, May 5-9, 2024. , May-2024
Abstracts for Journals and Proceedings	Christopher M, Jalil J, Weinreb RN, Laurie SS, Macias BR, and Huang AS. "Predicting spaceflight associated neuro-ocular syndrome using artificial intelligence." 2024 ARVO Annual Meeting, Seattle, Washington, May 5-9, 2024. Abstracts. 2024 ARVO Annual Meeting, Seattle, Washington, May 5-9, 2024. , May-2024
Articles in Peer-reviewed Journals	Moestl S, De Boni L, Hoenemann J-N, Kramer T, Schmitz J, Pesta D, Frett T, Bohmeier M, Frings-Meuthen P, Ewald AC, Nitsche A, Loehr P, Noppe A, Klischies N, Huang AS, Laurie SS, Marshall-Goebel K, Macias BR, Tank J, Jordan J, Mulder E. "Mechanical countermeasures for spaceflight-associated neuro-ocular syndrome during 30-days of head-down tilt bedrest: Design, implementation, and tolerability." Front Physiol. 2025 Feb 23;16:1530783. https://doi.org/10.3389/fphys.2025.1530783 ; PMID: 40066284; PMCID: PMC11891356 , Feb-2025